

Machine Learning Numerical Question Bank

50 Most Probable Exam-Style Questions with Full Computation, Formulas, Parameters, and Interpretations

Prepared in the style of the provided Sample_Numericals.pdf: questions are written as exam scenarios using phrases such as "Consider", "Suppose", "Given", and "Compute". Each solution includes the formula, parameter values used, step-by-step calculation, final answer, and interpretation wherever useful.

Primary numerical families covered: Gini, entropy, information gain, binning, normalization, confusion matrix metrics, rule coverage/accuracy, kNN, k-means, hierarchical clustering, and association rules.

How to write these in exam

1. Start by writing the given data clearly.
2. Write the formula before substitution.
3. Substitute values step by step.
4. Show intermediate values, not only the final result.
5. Box the final answer and add one short interpretation line.
6. For clustering, always mention the parameter used: k, initial centroids, linkage, distance metric, epsilon, MinPts, or bin count as applicable.

Formula Recap

Topic	Formula / Rule	Main parameter
Gini index	$\text{Gini}(D) = 1 - \sum p_i^2$	Class probabilities p_i
Entropy	$\text{Entropy}(D) = - \sum (p_i \log_2 p_i)$	Class probabilities p_i
Information gain	$\text{IG}(D,A) = \text{Entropy}(D) - \text{weighted child entropy}$	Split attribute A
Equal-width binning	$\text{width} = (\max - \min) / \text{number of bins}$	Number of bins k
Equal-frequency binning	$\text{records per bin} = n / \text{number of bins}$	Number of records n and bins k
Min-max normalization	$x_{\text{prime}} = (x - \min) / (\max - \min)$	Target range, usually [0,1]
Z-score	$z = (x - \text{mean}) / \text{std}$	Mean and standard deviation
Decimal scaling	$x_{\text{prime}} = x / 10^j$	Power j
Precision	$\text{TP} / (\text{TP} + \text{FP})$	TP, FP
Recall	$\text{TP} / (\text{TP} + \text{FN})$	TP, FN
F1-score	$2\text{PR} / (\text{P} + \text{R})$	Precision P and Recall R
Rule coverage	$\text{covered records} / \text{total records}$	Total records
Rule accuracy	$\text{correct covered records} / \text{covered records}$	Correct covered records
Euclidean distance	$\sqrt{\sum ((x_j - y_j)^2)}$	Distance metric
k-means SSE	$\text{sum over clusters and points } \ x - \text{centroid}\ ^2$	k and centroids
Support	$\text{count}(X) / \text{total transactions}$	Total transactions
Confidence	$\text{support}(X \cup Y) / \text{support}(X)$	Rule $X \rightarrow Y$
Lift	$\text{confidence}(X \rightarrow Y) / \text{support}(Y)$	Rule consequent Y

Question 1

Topic focus: Balanced binary Gini index

Question:

Consider a binary classification problem where the dataset contains 60 positive examples and 60 negative examples. Compute the Gini index for this dataset and briefly interpret the value.

Parameters used:

- Positive count = 60
- Negative count = 60

- Total records = 120
- Impurity measure used = Gini index

Detailed answer:

$$\text{Gini}(D) = 1 - (p_{\text{positive}}^2 + p_{\text{negative}}^2)$$

$$p_{\text{positive}} = 60/120 = 0.5$$

$$p_{\text{negative}} = 60/120 = 0.5$$

$$\text{Gini} = 1 - (0.5^2 + 0.5^2) = 1 - (0.25 + 0.25) = 0.5$$

Final answer: Gini index = 0.5

Interpretation: For a binary node, 0.5 is maximum impurity. The classes are perfectly balanced, so the node is highly mixed.

Question 2

Topic focus: Imbalanced binary Gini index

Question:

Suppose a binary classification dataset contains 80 positive examples and 20 negative examples. Compute the Gini index and interpret whether the dataset is more pure or more impure.

Parameters used:

- Positive count = 80
- Negative count = 20
- Total records = 100

Detailed answer:

$$\text{Gini} = 1 - (p_{\text{positive}}^2 + p_{\text{negative}}^2)$$

$$p_{\text{positive}} = 80/100 = 0.8$$

$$p_{\text{negative}} = 20/100 = 0.2$$

$$\text{Gini} = 1 - (0.8^2 + 0.2^2) = 1 - (0.64 + 0.04) = 0.32$$

Final answer: Gini index = 0.32

Interpretation: The node is impure because both classes exist, but it is less impure than a 50-50 node because positive class dominates.

Question 3

Topic focus: Three-class Gini index

Question:

Consider a node with 40 records of Class A, 30 records of Class B, and 30 records of Class C. Compute the Gini index for this node.

Parameters used:

- Class A = 40
- Class B = 30
- Class C = 30
- Total = 100

Detailed answer:

$$\text{Gini} = 1 - (p_A^2 + p_B^2 + p_C^2)$$

$$p_A = 40/100 = 0.4, p_B = 30/100 = 0.3, p_C = 30/100 = 0.3$$

$$\text{Gini} = 1 - (0.4^2 + 0.3^2 + 0.3^2)$$

$$\text{Gini} = 1 - (0.16 + 0.09 + 0.09) = 1 - 0.34 = 0.66$$

Final answer: Gini index = 0.66

Interpretation: The node is quite impure because records are distributed across three classes.

Question 4

Topic focus: Pure node Gini index

Question:

A decision tree node contains 100 records, and all records belong to Class A. Compute the Gini index and interpret the result.

Parameters used:

- Class A = 100
- Other classes = 0
- Total = 100

Detailed answer:

$$\text{Gini} = 1 - \sum(p_i^2)$$

$$p_A = 100/100 = 1$$

$$\text{Gini} = 1 - (1^2) = 0$$

Final answer: Gini index = 0

Interpretation: Gini = 0 means the node is perfectly pure. No further split is needed from an impurity point of view.

Question 5

Topic focus: Weighted Gini after a split

Question:

A parent node has 100 records. A candidate split creates a left child with 30 positive and 10 negative records, and a right child with 20 positive and 40 negative records. Compute the Gini index for each child and the weighted Gini of the split.

Parameters used:

- Left child: 30P, 10N, total 40
- Right child: 20P, 40N, total 60
- Split quality measure = weighted Gini

Detailed answer:

$$\text{Gini}(\text{child}) = 1 - (p_{\text{positive}}^2 + p_{\text{negative}}^2)$$

$$\text{Left child: } p_{\text{positive}} = 30/40 = 0.75, p_{\text{negative}} = 10/40 = 0.25$$

$$\text{Gini}_{\text{left}} = 1 - (0.75^2 + 0.25^2) = 1 - (0.5625 + 0.0625) = 0.375$$

$$\text{Right child: } p_{\text{positive}} = 20/60 = 0.333, p_{\text{negative}} = 40/60 = 0.667$$

$$\text{Gini}_{\text{right}} = 1 - (0.333^2 + 0.667^2) = 1 - (0.111 + 0.444) = 0.445$$

$$\text{Weighted Gini} = (40/100) * \text{Gini}_{\text{left}} + (60/100) * \text{Gini}_{\text{right}}$$

$$\text{Weighted Gini} = 0.4 * 0.375 + 0.6 * 0.445 = 0.150 + 0.267 = 0.417$$

Final answer: Weighted Gini of split = 0.417

Interpretation: A lower weighted Gini means a better split. This split reduces impurity only moderately.

Question 6

Topic focus: Binary entropy

Question:

Consider a node with 9 positive records and 5 negative records. Compute the entropy of the node.

Parameters used:

- Positive = 9
- Negative = 5
- Total = 14
- Log base = 2

Detailed answer:

$$\text{Entropy}(D) = -p_{\text{positive}} \log_2(p_{\text{positive}}) - p_{\text{negative}} \log_2(p_{\text{negative}})$$

$$p_{\text{positive}} = 9/14 = 0.643$$

$$p_{\text{negative}} = 5/14 = 0.357$$

$$\text{Entropy} = -0.643 \log_2(0.643) - 0.357 \log_2(0.357)$$

$$\log_2(0.643) = -0.637, \log_2(0.357) = -1.485$$

$$\text{Entropy} = -0.643 \cdot (-0.637) - 0.357 \cdot (-1.485) = 0.410 + 0.530 = 0.940$$

Final answer: Entropy approximately 0.940 bits

Interpretation: The node has high uncertainty because both classes are present in meaningful quantities.

Question 7

Topic focus: Balanced binary entropy

Question:

Suppose a binary node contains 50 positive and 50 negative records. Compute the entropy and interpret the value.

Parameters used:

- Positive = 50
- Negative = 50
- Total = 100

Detailed answer:

$$\text{Entropy} = -p_{\text{positive}} \log_2(p_{\text{positive}}) - p_{\text{negative}} \log_2(p_{\text{negative}})$$

$$p_{\text{positive}} = 0.5, p_{\text{negative}} = 0.5$$

$$\text{Entropy} = -0.5 \log_2(0.5) - 0.5 \log_2(0.5)$$

$$\log_2(0.5) = -1$$

$$\text{Entropy} = 0.5 + 0.5 = 1$$

Final answer: Entropy = 1 bit

Interpretation: For binary classification, entropy = 1 is maximum uncertainty. The node is perfectly mixed.

Question 8

Topic focus: Pure entropy

Question:

A node contains 20 records of Class A and 0 records of Class B. Compute entropy.

Parameters used:

- Class A = 20
- Class B = 0
- Total = 20

Detailed answer:

$$\text{Entropy} = -\sum(p_i \log_2 p_i)$$

$$p_A = 20/20 = 1, p_B = 0/20 = 0$$

$$\text{Entropy} = -(1 \log_2(1)) = 0 \text{ because } \log_2(1) = 0. \text{ The zero-probability class contributes 0.}$$

Final answer: Entropy = 0

Interpretation: The node is pure. There is no uncertainty about class label.

Question 9

Topic focus: Information gain for one split

Question:

A parent node has 10 records: 6 positive and 4 negative. Attribute A splits it into Child 1 with 4 positive and 1 negative, and Child 2 with 2 positive and 3 negative. Compute parent entropy, weighted child entropy, and information gain.

Parameters used:

- Parent = 6P, 4N
- Child 1 = 4P, 1N
- Child 2 = 2P, 3N
- Split attribute = A

Detailed answer:

$IG(D,A) = \text{Entropy}(\text{parent}) - \text{weighted child entropy}$

Parent: $pP=6/10=0.6$, $pN=4/10=0.4$

$\text{Entropy}(\text{parent}) = -0.6\log_2(0.6) - 0.4\log_2(0.4) = 0.971$

Child 1: $pP=4/5=0.8$, $pN=1/5=0.2$

$\text{Entropy}(C1) = -0.8\log_2(0.8) - 0.2\log_2(0.2) = 0.722$

Child 2: $pP=2/5=0.4$, $pN=3/5=0.6$

$\text{Entropy}(C2) = -0.4\log_2(0.4) - 0.6\log_2(0.6) = 0.971$

Weighted child entropy = $(5/10)*0.722 + (5/10)*0.971 = 0.8465$

Information gain = $0.971 - 0.8465 = 0.1245$

Final answer: Information gain approximately 0.125

Interpretation: The split reduces uncertainty, but the reduction is modest.

Question 10

Topic focus: Choosing the best split using information gain

Question:

A node has 12 records: 7 positive and 5 negative. Attribute A creates two groups: A1 = 4P,2N and A2 = 3P,3N. Attribute B creates two groups: B1 = 5P,1N and B2 = 2P,4N. Compute information gain for both attributes and choose the better split.

Parameters used:

- Parent = 7P,5N
- Attribute A split: (4P,2N), (3P,3N)
- Attribute B split: (5P,1N), (2P,4N)
- Criterion = highest information gain

Detailed answer:

Parent entropy: $pP=7/12=0.583$, $pN=5/12=0.417$

$\text{Entropy}(\text{parent}) = -0.583\log_2(0.583) - 0.417\log_2(0.417) = 0.980$

For Attribute A

A1 entropy for 4P,2N: $pP=0.667$, $pN=0.333 \rightarrow \text{Entropy}=0.918$

A2 entropy for 3P,3N: $pP=0.5$, $pN=0.5 \rightarrow \text{Entropy}=1.000$

Weighted entropy A = $(6/12)*0.918 + (6/12)*1.000 = 0.959$

$IG(A) = 0.980 - 0.959 = 0.021$

For Attribute B

B1 entropy for 5P,1N: $pP=0.833$, $pN=0.167 \rightarrow \text{Entropy}=0.650$

B2 entropy for 2P,4N: $pP=0.333$, $pN=0.667 \rightarrow \text{Entropy}=0.918$

Weighted entropy B = $(6/12)*0.650 + (6/12)*0.918 = 0.784$

$IG(B) = 0.980 - 0.784 = 0.196$

Final answer: Attribute B is better because $IG(B)=0.196 > IG(A)=0.021$.

Interpretation: Decision trees choose the split that gives higher impurity reduction.

Question 11

Topic focus: Equal-frequency binning of salary records

Question:

Suppose the following salary records are already sorted: 12000, 15000, 18000, 21000, 25000, 31000, 36000, 42000, 48000, 60000, 66000, 72000. Partition them into 3 bins using equal-frequency binning.

Parameters used:

- Number of records $n = 12$
- Number of bins $k = 3$
- Binning method = equal-frequency/equal-depth

Detailed answer:

$$\text{Records per bin} = n/k = 12/3 = 4$$

Place 4 records in each bin because equal-frequency binning balances the count per bin.

Bin 1: 12000, 15000, 18000, 21000

Bin 2: 25000, 31000, 36000, 42000

Bin 3: 48000, 60000, 66000, 72000

Final answer: 3 equal-frequency bins, each containing 4 records.

Interpretation: This method balances record count, but bin widths may be unequal.

Question 12

Topic focus: Equal-width binning of salary records

Question:

Using the same salary records from the previous question, partition the values into 3 bins using equal-width binning and show the bin ranges.

Parameters used:

- Minimum value = 12000
- Maximum value = 72000
- Number of bins $k = 3$

Detailed answer:

$$\text{Width} = (\text{max} - \text{min})/k$$

$$\text{Width} = (72000 - 12000)/3 = 60000/3 = 20000$$

Bin 1 range: 12000 to 32000

Values: 12000, 15000, 18000, 21000, 25000, 31000

Bin 2 range: >32000 to 52000

Values: 36000, 42000, 48000

Bin 3 range: >52000 to 72000

Values: 60000, 66000, 72000

Final answer: Equal-width bin size = 20000.

Interpretation: Equal-width binning balances interval size, not number of records.

Question 13

Topic focus: Equal-frequency and equal-width binning for ages

Question:

Consider sorted ages: 18, 21, 23, 25, 29, 34, 38, 42, 48, 55, 60, 70. Create 4 bins using equal-frequency and equal-width binning.

Parameters used:

- $n = 12$ records
- $k = 4$ bins
- $\text{min} = 18$

- max = 70

Detailed answer:

Equal-frequency

Records per bin = $12/4 = 3$

Bin 1: 18, 21, 23
Bin 2: 25, 29, 34
Bin 3: 38, 42, 48
Bin 4: 55, 60, 70

Equal-width

Width = $(70 - 18)/4 = 52/4 = 13$

Bin 1: 18 to 31 -> 18, 21, 23, 25, 29
Bin 2: >31 to 44 -> 34, 38, 42
Bin 3: >44 to 57 -> 48, 55
Bin 4: >57 to 70 -> 60, 70

Final answer: Equal-frequency has 3 records per bin; equal-width has interval width 13.

Interpretation: The two methods produce different bins because one uses count and the other uses numeric range.

Question 14

Topic focus: Equal-frequency binning with uneven records

Question:

Suppose sorted values are: 5, 7, 8, 10, 12, 14, 18, 21, 25, 30. Partition the data into 3 equal-frequency bins. Explain how you handle uneven bin size.

Parameters used:

- n = 10 records
- k = 3 bins
- Method = equal-frequency

Detailed answer:

Since $10/3 = 3.33$, equal bin sizes are not possible. A common exam-style solution is to make bins as balanced as possible, such as 4,3,3 records.

Bin 1: 5, 7, 8, 10 (4 records)
Bin 2: 12, 14, 18 (3 records)
Bin 3: 21, 25, 30 (3 records)

Final answer: One valid equal-frequency partition is 4-3-3 records.

Interpretation: When n is not divisible by k, bins should be as balanced as possible.

Question 15

Topic focus: Equal-width binning affected by outliers

Question:

Given income values 10000, 12000, 15000, 16000, 18000, 20000, 22000, 80000, 90000, perform equal-width binning into 3 bins and explain why this method may be affected by outliers.

Parameters used:

- min = 10000
- max = 90000
- k = 3 bins

Detailed answer:

Width = $(90000 - 10000)/3 = 80000/3 = 26666.67$

Bin 1: 10000 to 36666.67 -> 10000, 12000, 15000, 16000, 18000, 20000, 22000
Bin 2: >36666.67 to 63333.34 -> no records
Bin 3: >63333.34 to 90000 -> 80000, 90000

Final answer: Equal-width bins are highly uneven for this dataset.

Interpretation: The outliers 80000 and 90000 stretch the range, creating an empty middle bin. Equal-frequency may be better for skewed data.

Question 16

Topic focus: Min-max normalization to [0,1]

Question:

Consider the dataset 10, 20, 30, 40, 50. Perform min-max normalization to the range [0,1]. Also calculate the normalized value for 20.

Parameters used:

- min = 10
- max = 50
- Target range = [0,1]

Detailed answer:

$$x_prime = (x - min)/(max - min)$$

For 10: $(10-10)/(50-10)=0/40=0$

For 20: $(20-10)/40=10/40=0.25$

For 30: $20/40=0.50$

For 40: $30/40=0.75$

For 50: $40/40=1.00$

Final answer: Normalized values = 0.00, 0.25, 0.50, 0.75, 1.00. Normalized value for 20 = 0.25.

Interpretation: Min-max normalization preserves order and rescales all values into a fixed range.

Question 17

Topic focus: Min-max normalization to custom range [-1,1]

Question:

Suppose values are 5, 10, 15, 20, 25. Normalize them to the range [-1,1] using min-max normalization.

Parameters used:

- min = 5
- max = 25
- New minimum a = -1
- New maximum b = 1

Detailed answer:

$$x_prime = ((x - min)/(max - min)) * (b - a) + a$$

Here max-min = 25-5 = 20 and b-a = 1-(-1)=2.

For 5: $((5-5)/20)*2 - 1 = -1$

For 10: $(5/20)*2 - 1 = 0.5 - 1 = -0.5$

For 15: $(10/20)*2 - 1 = 1 - 1 = 0$

For 20: $(15/20)*2 - 1 = 1.5 - 1 = 0.5$

For 25: $(20/20)*2 - 1 = 1$

Final answer: Normalized values = -1, -0.5, 0, 0.5, 1.

Interpretation: Custom range normalization is useful when an algorithm expects inputs centered around zero.

Question 18

Topic focus: Z-score normalization

Question:

Consider the dataset 10, 20, 30, 40, 50. Given mean = 30 and standard deviation = 14.14, perform z-score normalization.

Parameters used:

- Mean = 30
- Standard deviation = 14.14
- Formula = $z = (x - \text{mean})/\text{std}$

Detailed answer:

For 10: $(10-30)/14.14 = -20/14.14 = -1.41$

For 20: $(20-30)/14.14 = -10/14.14 = -0.71$

For 30: $(30-30)/14.14 = 0$

For 40: $(40-30)/14.14 = 10/14.14 = 0.71$

For 50: $(50-30)/14.14 = 20/14.14 = 1.41$

Final answer: Z-score values = -1.41, -0.71, 0.00, 0.71, 1.41.

Interpretation: Z-score tells how many standard deviations a value is from the mean.

Question 19

Topic focus: Decimal scaling normalization

Question:

Normalize the values 100, 200, 350, 500, 900 using decimal scaling.

Parameters used:

- Maximum absolute value = 900
- Choose j such that $\max(|x|)/10^j < 1$
- $j = 3$ because $900/1000 = 0.9$

Detailed answer:

$$x_{\text{prime}} = x / 10^j$$

100 -> $100/1000 = 0.100$

200 -> 0.200

350 -> 0.350

500 -> 0.500

900 -> 0.900

Final answer: Normalized values = 0.100, 0.200, 0.350, 0.500, 0.900.

Interpretation: Decimal scaling reduces magnitude by moving the decimal point.

Question 20

Topic focus: Comparison of normalization methods

Question:

Given the values 20, 40, 60, 80, perform min-max normalization, z-score normalization, and decimal scaling. Explain which method is useful for distance-based algorithms.

Parameters used:

- Values = 20,40,60,80
- For min-max: min=20, max=80
- For z-score: mean=50, std approximately 22.36
- For decimal scaling: $j=2$

Detailed answer:**Min-max normalization**

20 -> 0, 40 -> $(20/60)=0.333$, 60 -> $(40/60)=0.667$, 80 -> 1

Z-score normalization

$$20 \rightarrow (20-50)/22.36 = -1.34$$

$$40 \rightarrow -0.45, 60 \rightarrow 0.45, 80 \rightarrow 1.34$$

Decimal scaling

$$20/100=0.20, 40/100=0.40, 60/100=0.60, 80/100=0.80$$

Final answer: Min-max: 0,0.333,0.667,1; z-score: -1.34,-0.45,0.45,1.34; decimal: 0.20,0.40,0.60,0.80.

Interpretation: Distance-based algorithms such as kNN and k-means need scaling so one large-range feature does not dominate distance.

Question 21

Topic focus: Precision and recall for spam classifier

Question:

Suppose a binary classification model predicts whether an email is Spam or Not Spam. The confusion matrix is: Actual Spam predicted Spam = 45, Actual Spam predicted Not Spam = 15, Actual Not Spam predicted Spam = 5, Actual Not Spam predicted Not Spam = 35. Calculate precision and recall for the Spam class.

Parameters used:

- Positive class = Spam
- TP = 45
- FN = 15
- FP = 5
- TN = 35

Detailed answer:

$$\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$$

$$\text{Precision} = 45/(45+5) = 45/50 = 0.90$$

$$\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$$

$$\text{Recall} = 45/(45+15) = 45/60 = 0.75$$

Final answer: Precision = 90%; Recall = 75%.

Interpretation: Precision tells how many predicted spam emails were truly spam. Recall tells how many actual spam emails were detected.

Question 22

Topic focus: Accuracy, precision, recall, and F1-score

Question:

A disease classifier gives the following confusion matrix: Actual Positive predicted Positive = 36, Actual Positive predicted Negative = 14, Actual Negative predicted Positive = 6, Actual Negative predicted Negative = 44. Compute accuracy, precision, recall, and F1-score.

Parameters used:

- TP=36
- FN=14
- FP=6
- TN=44
- Positive class = Disease present

Detailed answer:

$$\text{Total} = \text{TP}+\text{FN}+\text{FP}+\text{TN} = 36+14+6+44 = 100$$

$$\text{Accuracy} = (\text{TP}+\text{TN})/\text{Total} = (36+44)/100 = 80/100 = 0.80$$

$$\text{Precision} = \text{TP}/(\text{TP}+\text{FP}) = 36/(36+6) = 36/42 = 0.857$$

$$\text{Recall} = \text{TP}/(\text{TP}+\text{FN}) = 36/(36+14) = 36/50 = 0.72$$

$$\text{F1} = 2\text{PR}/(\text{P}+\text{R}) = 2*(0.857*0.72)/(0.857+0.72) = 1.234/(1.577) = 0.782$$

Final answer: Accuracy = 80%, Precision = 85.7%, Recall = 72%, F1 = 78.2%.

Interpretation: F1 combines precision and recall, useful when both false positives and false negatives matter.

Question 23

Topic focus: Medical classifier with specificity

Question:

A disease classifier produces TP = 80, FP = 20, FN = 10, and TN = 90. Compute accuracy, precision, recall, specificity, and F1-score.

Parameters used:

- TP=80
- FP=20
- FN=10
- TN=90

Detailed answer:

Total = $80+20+10+90 = 200$

Accuracy = $(TP+TN)/\text{Total} = (80+90)/200 = 170/200 = 0.85$

Precision = $TP/(TP+FP) = 80/(80+20)=0.80$

Recall = $TP/(TP+FN) = 80/(80+10)=80/90=0.889$

Specificity = $TN/(TN+FP) = 90/(90+20)=90/110=0.818$

F1 = $2PR/(P+R) = 2*(0.80*0.889)/(0.80+0.889)=1.422/1.689=0.842$

Final answer: Accuracy=85%, Precision=80%, Recall=88.9%, Specificity=81.8%, F1=84.2%.

Interpretation: High recall means the model detects most actual disease cases.

Question 24

Topic focus: Imbalanced fraud classification

Question:

A fraud classifier gives TP = 15, FP = 5, FN = 35, and TN = 945. Compute accuracy and recall. Explain why accuracy can be misleading.

Parameters used:

- TP=15
- FP=5
- FN=35
- TN=945
- Positive class = Fraud

Detailed answer:

Total = $15+5+35+945 = 1000$

Accuracy = $(TP+TN)/\text{Total} = (15+945)/1000 = 960/1000 = 0.96$

Recall = $TP/(TP+FN) = 15/(15+35) = 15/50 = 0.30$

Final answer: Accuracy = 96%, Recall = 30%.

Interpretation: Accuracy looks high because most transactions are non-fraud, but recall shows the model misses 70% of fraud cases.

Question 25

Topic focus: Choosing a classifier based on precision and recall

Question:

Classifier A has precision = 0.90 and recall = 0.50. Classifier B has precision = 0.70 and recall = 0.80. Which classifier is better for disease detection and why?

Parameters used:

- Task = disease detection
- Metric focus = recall because missing disease is costly

Detailed answer:

Disease detection usually prioritizes recall because false negatives are dangerous.

Classifier A: high precision, but recall = 0.50 means it misses 50% of actual disease cases.

Classifier B: lower precision, but recall = 0.80 means it detects 80% of actual disease cases.

Therefore, Classifier B is usually better for screening/disease detection.

Final answer: Choose Classifier B for disease detection.

Interpretation: In high-risk medical screening, missing positive cases is usually worse than raising some false alarms.

Question 26

Topic focus: Basic rule coverage and accuracy

Question:

A dataset contains 200 records. A rule covers 40 records. Among the 40 covered records, 32 are correctly classified. Compute the rule coverage and rule accuracy.

Parameters used:

- Total records = 200
- Records covered by rule = 40
- Correct covered records = 32

Detailed answer:

Coverage = records covered / total records

Coverage = $40/200 = 0.20 = 20\%$

Rule accuracy = correct covered records / records covered

Rule accuracy = $32/40 = 0.80 = 80\%$

Final answer: Coverage = 20%; Rule accuracy = 80%.

Interpretation: The rule is reliable on covered cases but covers only one-fifth of the dataset.

Question 27

Topic focus: Multiple rule coverage and accuracy

Question:

A dataset has 100 records. Four rules cover and correctly classify the following records: R1 covers 10 and gets 9 correct, R2 covers 20 and gets 14 correct, R3 covers 25 and gets 15 correct, and R4 covers 5 and gets 5 correct. Compute coverage and accuracy for each rule and rank the rules.

Parameters used:

- Total records = 100
- Rule quality metrics = coverage and accuracy

Detailed answer:

Rule	Covered	Correct	Coverage	Accuracy
R1	10	9	$10/100 = 0.10$	$9/10 = 0.90$
R2	20	14	$20/100 = 0.20$	$14/20 = 0.70$
R3	25	15	$25/100 = 0.25$	$15/25 = 0.60$
R4	5	5	$5/100 = 0.05$	$5/5 = 1.00$

If ranking only by accuracy: $R4 > R1 > R2 > R3$.

If considering both coverage and accuracy, R1 is often more useful than R4 because R4 is perfect but covers very few records.

Final answer: R4 has highest accuracy, but R1 is the best balanced rule.

Interpretation: A good rule should have both high accuracy and meaningful coverage.

Question 28

Topic focus: Five-rule quality assessment

Question:

In a rule-based classifier, total records = 100. R1 covers 15 records and 12 are correct. R2 covers 25 and 16 are correct. R3 covers 10 and 9 are correct. R4 covers 30 and 18 are correct. R5 covers 20 and 14 are correct. Compute coverage and accuracy for each rule and assess rule quality.

Parameters used:

- Total records = 100
- Metrics = coverage, accuracy

Detailed answer:

Rule	Covered	Correct	Coverage	Accuracy
R1	15	12	0.15	0.80
R2	25	16	0.25	0.64
R3	10	9	0.10	0.90
R4	30	18	0.30	0.60
R5	20	14	0.20	0.70

R3 has the highest accuracy, but low coverage. R1 has good accuracy with moderate coverage. R4 has highest coverage but low accuracy.

Best quality depends on goal: choose R3 for reliability, R1 for balanced usefulness, and avoid R4 unless coverage is more important than correctness.

Final answer: R1 and R3 are strongest; R4 needs refinement.

Interpretation: Rule quality must be judged using both coverage and accuracy together.

Question 29

Topic focus: High accuracy but low coverage rule

Question:

A rule covers 5 out of 100 records and correctly classifies all 5. Compute coverage and accuracy. Is it a strong rule?

Parameters used:

- Total = 100
- Covered = 5
- Correct = 5

Detailed answer:

Coverage = $5/100 = 0.05 = 5\%$

Accuracy = $5/5 = 1.00 = 100\%$

Final answer: Coverage = 5%; Accuracy = 100%.

Interpretation: The rule is very accurate but not very general. It may be too specific and may not help classify many records.

Question 30

Topic focus: High coverage but low accuracy rule

Question:

A rule covers 80 out of 100 records but correctly classifies only 48 of them. Compute coverage and accuracy. Is the rule useful?

Parameters used:

- Total = 100

- Covered = 80

- Correct = 48

Detailed answer:

Coverage = $80/100 = 0.80 = 80\%$

Accuracy = $48/80 = 0.60 = 60\%$

Final answer: Coverage = 80%; Accuracy = 60%.

Interpretation: The rule is general but not reliable. It should be refined because it makes many mistakes.

Question 31

Topic focus: kNN classification using Euclidean distance

Question:

A new point $X = (4,4)$ must be classified using kNN with $k = 3$. Training data: $(2,3) \rightarrow A$, $(3,5) \rightarrow A$, $(6,5) \rightarrow B$, $(7,7) \rightarrow B$, $(4,2) \rightarrow A$. Use Euclidean distance and predict the class.

Parameters used:

- $k = 3$

- Distance metric = Euclidean

- Task = classification

- Voting = majority vote

Detailed answer:

$$d(X,P) = \sqrt{(x_1-p_1)^2 + (x_2-p_2)^2}$$

Point	Class	Distance from (4,4)
(2,3)	A	$\sqrt{2^2+1^2}=2.236$
(3,5)	A	$\sqrt{1^2+(-1)^2}=1.414$
(6,5)	B	$\sqrt{(-2)^2+(-1)^2}=2.236$
(7,7)	B	$\sqrt{(-3)^2+(-3)^2}=4.243$
(4,2)	A	$\sqrt{0^2+2^2}=2.000$

The 3 nearest neighbors are $(3,5) \rightarrow A$, $(4,2) \rightarrow A$, and one of the points at distance 2.236. If tie is resolved by listed order, the third is $(2,3) \rightarrow A$.

Majority among nearest 3 = A, A, A.

Final answer: Predicted class = A.

Interpretation: kNN assigns the query to the majority class among its nearest neighbors.

Question 32

Topic focus: Effect of different k values in kNN

Question:

Using the same data as the previous question, classify $X=(4,4)$ using $k=1$, $k=3$, and $k=5$. Explain how the result changes.

Parameters used:

- Distance metric = Euclidean

- Task = classification

- k values = 1, 3, 5

Detailed answer:

Distances sorted approximately:

1.414 $\rightarrow$ (3,5) Class A
2.000 $\rightarrow$ (4,2) Class A
2.236 $\rightarrow$ (2,3) Class A
2.236 $\rightarrow$ (6,5) Class B
4.243 $\rightarrow$ (7,7) Class B

$k=1$: nearest point is Class A, so prediction = A.

k=3: nearest three are A, A, A, so prediction = A.

k=5: labels are A, A, A, B, B, so majority = A.

Final answer: For k=1,3,5, predicted class = A.

Interpretation: In other datasets, changing k may change the prediction. Small k is sensitive; large k is smoother.

Question 33

Topic focus: kNN regression using average

Question:

Suppose a regression dataset has training points (1,10), (2,12), (3,14), (8,30), and (9,32). Predict y for x = 2.5 using k = 3 and simple average of neighbor target values.

Parameters used:

- Task = regression
- k = 3
- Distance = absolute distance in 1D
- Aggregation = simple average

Detailed answer:

Training x	y	Distance from x=2.5
1	10	1.5
2	12	0.5
3	14	0.5
8	30	5.5
9	32	6.5

The 3 nearest neighbors are x=2 (y=12), x=3 (y=14), and x=1 (y=10).

Predicted y = $(12+14+10)/3 = 36/3 = 12$

Final answer: Predicted y = 12.

Interpretation: For regression, kNN predicts a numeric value by averaging nearby target values.

Question 34

Topic focus: Distance-weighted kNN regression

Question:

Using the same dataset, predict y for x = 2.5 using k = 3 and inverse-distance weighted average. Use weights $w = 1/d$.

Parameters used:

- k = 3
- Neighbors: x=2, x=3, x=1
- Weights = 1/distance
- Distances = 0.5, 0.5, 1.5

Detailed answer:

Neighbor values: x=2 -> y=12, distance=0.5, weight=1/0.5=2

x=3 -> y=14, distance=0.5, weight=2

x=1 -> y=10, distance=1.5, weight=0.667

$$\text{Weighted prediction} = \frac{\sum(w_i y_i)}{\sum(w_i)}$$

$$\text{Numerator} = 2*12 + 2*14 + 0.667*10 = 24 + 28 + 6.67 = 58.67$$

$$\text{Denominator} = 2 + 2 + 0.667 = 4.667$$

$$\text{Prediction} = 58.67/4.667 = 12.57$$

Final answer: Weighted kNN prediction approximately 12.57.

Interpretation: Closer neighbors have more influence than farther neighbors.

Question 35

Topic focus: Feature scaling effect in kNN

Question:

Two training customers are A(age=25, salary=30000) and B(age=30, salary=80000). Query customer Q has age=27 and salary=35000. Compute Euclidean distance before and after min-max scaling. Use age range 20 to 60 and salary range 20000 to 100000.

Parameters used:

- Distance metric = Euclidean
- Age min=20, max=60
- Salary min=20000, max=100000
- Scaling = min-max

Detailed answer:

Before scaling

$$d(Q,A)=\sqrt{((27-25)^2 + (35000-30000)^2)} = \sqrt{4 + 25,000,000} = 5000.0004$$

$$d(Q,B)=\sqrt{((27-30)^2 + (35000-80000)^2)} = \sqrt{9 + 2,025,000,000} = 45000.0001$$

Before scaling, A is closer. Salary dominates the distance.

After scaling

$$\text{Age scaled: } A=(25-20)/40=0.125, B=(30-20)/40=0.25, Q=(27-20)/40=0.175$$

$$\text{Salary scaled: } A=(30000-20000)/80000=0.125, B=(80000-20000)/80000=0.75, Q=(35000-20000)/80000=0.1875$$

$$d_scaled(Q,A)=\sqrt{((0.175-0.125)^2 + (0.1875-0.125)^2)}=\sqrt{0.0025+0.003906}=0.080$$

$$d_scaled(Q,B)=\sqrt{((0.175-0.25)^2 + (0.1875-0.75)^2)}=\sqrt{0.005625+0.316406}=0.568$$

Final answer: After scaling also A is closer, but now both age and salary contribute fairly.

Interpretation: Scaling prevents large-range features such as salary from overwhelming smaller-range features such as age.

Question 36

Topic focus: One iteration of k-means in 1D

Question:

Consider points 1, 2, 3, 10, 11, 12. Let $k = 2$ and initial centroids be 2 and 11. Perform one assignment step and one centroid update.

Parameters used:

- $k = 2$
- Initial centroid $C1=2$
- Initial centroid $C2=11$
- Distance = absolute distance

Detailed answer:

Point	Distance to $C1=2$	Distance to $C2=11$	Assigned cluster
1	1	10	C1
2	0	9	C1
3	1	8	C1
10	8	1	C2
11	9	0	C2
12	10	1	C2

$C1$ contains {1,2,3}. New $C1 = (1+2+3)/3 = 2$.

$C2$ contains {10,11,12}. New $C2 = (10+11+12)/3 = 11$.

Final answer: New centroids remain C1=2 and C2=11.

Interpretation: Since centroids did not change, k-means has converged for this initialization.

Question 37

Topic focus: k-means with 2D points

Question:

Consider points (1,2), (2,1), (2,2), (8,8), (9,8), (8,9). Let $k = 2$ with initial centroids $C1=(1,2)$ and $C2=(8,8)$. Assign each point to the nearest centroid and compute new centroids.

Parameters used:

- $k=2$
- Initial $C1=(1,2)$
- Initial $C2=(8,8)$
- Distance metric = Euclidean

Detailed answer:

Point	d to C1	d to C2	Cluster
(1,2)	0.00	9.22	C1
(2,1)	1.41	9.22	C1
(2,2)	1.00	8.49	C1
(8,8)	9.22	0.00	C2
(9,8)	10.00	1.00	C2
(8,9)	10.00	1.00	C2

$C1 = \{(1,2), (2,1), (2,2)\}$. New $C1 = ((1+2+2)/3, (2+1+2)/3) = (1.67, 1.67)$.

$C2 = \{(8,8), (9,8), (8,9)\}$. New $C2 = ((8+9+8)/3, (8+8+9)/3) = (8.33, 8.33)$.

Final answer: New centroids: $C1=(1.67, 1.67)$, $C2=(8.33, 8.33)$.

Interpretation: K-means updates each centroid using the mean of points assigned to that cluster.

Question 38

Topic focus: k-means SSE for final clusters

Question:

Using the final clusters from the previous question, compute the sum of squared errors (SSE).

Parameters used:

- Cluster 1 centroid = (1.67, 1.67)
- Cluster 2 centroid = (8.33, 8.33)
- SSE uses squared Euclidean distance

Detailed answer:

Cluster 1 SSE

For (1,2): $(1-1.67)^2 + (2-1.67)^2 = 0.4489 + 0.1089 = 0.5578$

For (2,1): $0.1089 + 0.4489 = 0.5578$

For (2,2): $0.1089 + 0.1089 = 0.2178$

$SSE1 = 0.5578 + 0.5578 + 0.2178 = 1.3334$

Cluster 2 SSE

Symmetric values: (8,8), (9,8), (8,9) around (8.33, 8.33) give SSE2 approximately 1.3334.

Total SSE = $1.3334 + 1.3334 = 2.6668$

Final answer: Total SSE approximately 2.67.

Interpretation: SSE measures compactness of clusters; lower SSE means points are closer to their centroids.

Question 39

Topic focus: k-means with 3 clusters - first iteration

Question:

Suppose eight 2D points are A1(10,2), A2(5,2), A3(8,4), A4(5,8), A5(5,7), A6(6,4), A7(1,2), and A8(9,4). Use $k = 3$ with initial centers A1, A4, and A7. Perform the first k-means assignment and compute new centroids.

Parameters used:

- $k=3$
- Initial $C1=A1=(10,2)$
- Initial $C2=A4=(5,8)$
- Initial $C3=A7=(1,2)$
- Distance metric = Euclidean

Detailed answer:

Point	d to C1	d to C2	d to C3	Cluster
A1(10,2)	0.00	7.81	9.00	C1
A2(5,2)	5.00	6.00	4.00	C3
A3(8,4)	2.83	5.00	7.28	C1
A4(5,8)	7.81	0.00	7.21	C2
A5(5,7)	7.07	1.00	6.40	C2
A6(6,4)	4.47	4.12	5.39	C2
A7(1,2)	9.00	7.21	0.00	C3
A8(9,4)	2.24	5.66	8.25	C1

$C1 = \{A1, A3, A8\}$; new $C1 = ((10+8+9)/3, (2+4+4)/3) = (9, 3.33)$.

$C2 = \{A4, A5, A6\}$; new $C2 = ((5+5+6)/3, (8+7+4)/3) = (5.33, 6.33)$.

$C3 = \{A2, A7\}$; new $C3 = ((5+1)/2, (2+2)/2) = (3, 2)$.

Final answer: New centers: $C1=(9,3.33)$, $C2=(5.33,6.33)$, $C3=(3,2)$.

Interpretation: The first iteration groups points around the nearest initial centers.

Question 40

Topic focus: k-means second iteration and convergence

Question:

Using the new centers from the previous question, perform the second assignment step and state whether the clusters change. Also compute final SSE.

Parameters used:

- Centers: $C1=(9,3.33)$, $C2=(5.33,6.33)$, $C3=(3,2)$
- Distance metric = Euclidean

Detailed answer:

Point	d to C1	d to C2	d to C3	Cluster
A1	1.67	6.37	7.00	C1
A2	4.22	4.35	2.00	C3
A3	1.20	3.54	5.39	C1
A4	6.15	1.70	6.32	C2
A5	5.43	0.75	5.39	C2
A6	3.07	2.43	3.61	C2
A7	8.11	6.13	2.00	C3
A8	0.67	4.35	6.32	C1

Assignments remain $C1=\{A1, A3, A8\}$, $C2=\{A4, A5, A6\}$, $C3=\{A2, A7\}$. Therefore the algorithm has converged.

SSE C1 around (9,3.33): $A1=2.78$, $A3=1.44$, $A8=0.44$ -> $SSE1=4.66$

SSE C2 around (5.33,6.33): $A4=2.89$, $A5=0.56$, $A6=5.89$ -> $SSE2=9.34$

SSE C3 around (3,2): $A2=4$, $A7=4$ -> $SSE3=8$

Total SSE = $4.66 + 9.34 + 8 = 22.00$

Final answer: Clusters do not change; final SSE approximately 22.00.

Interpretation: Convergence happens when assignments or centroids stop changing.

Question 41

Topic focus: Proximity matrix for hierarchical clustering

Question:

Construct the proximity matrix for the one-dimensional points 0, 4, 5, 20, 25, 39, 43, and 44 using absolute distance.

Parameters used:

- Distance metric = absolute distance $|x-y|$
- Data type = one-dimensional numeric points

Detailed answer:

For one-dimensional points, distance between x and y is $|x-y|$. For example, $d(4,5)=1$ and $d(20,25)=5$.

	0	4	5	20	25	39	43	44
0	0	4	5	20	25	39	43	44
4	4	0	1	16	21	35	39	40
5	5	1	0	15	20	34	38	39
20	20	16	15	0	5	19	23	24
25	25	21	20	5	0	14	18	19
39	39	35	34	19	14	0	4	5
43	43	39	38	23	18	4	0	1
44	44	40	39	24	19	5	1	0

Final answer: The above symmetric matrix is the proximity matrix.

Interpretation: The proximity matrix is used to find the closest clusters at each step of hierarchical clustering.

Question 42

Topic focus: Single linkage hierarchical clustering

Question:

Using the points 0, 4, 5, 20, 25, 39, 43, and 44, perform hierarchical agglomerative clustering using single linkage. Show the merge order.

Parameters used:

- Linkage = single/min linkage
- Cluster distance = minimum pairwise distance
- Distance metric = absolute distance

Detailed answer:

$d(A,B)$ = min distance between any point in cluster A and any point in cluster B

Start with each point as its own cluster.

Step 1: Merge 4 and 5 at distance 1 -> {4,5}
Step 2: Merge 43 and 44 at distance 1 -> {43,44}
Step 3: $d(\{4,5\},0)=\min(|4-0|,|5-0|)=\min(4,5)=4$ -> merge {0,4,5}
Step 4: $d(\{43,44\},39)=\min(4,5)=4$ -> merge {39,43,44}
Step 5: Merge 20 and 25 at distance 5 -> {20,25}
Step 6: $d(\{20,25\},\{39,43,44\})=\min(19,23,24,14,18,19)=14$ -> merge
Step 7: $d(\{0,4,5\},\{20,25,39,43,44\})=\min(20,25,39,43,44,16,21,35,39,40,15,20,34,38,39)=15$ -> final merge

Final answer: Merge distances: 1, 1, 4, 4, 5, 14, 15.

Interpretation: Single linkage can create chaining because it only considers the nearest pair between clusters.

Question 43

Topic focus: Complete linkage hierarchical clustering

Question:

Using the same points, perform hierarchical agglomerative clustering using complete linkage. Show the merge order.

Parameters used:

- Linkage = complete/max linkage
- Cluster distance = maximum pairwise distance

Detailed answer:

$d(A,B)$ = max distance between any point in cluster A and any point in cluster B

Step 1: Merge 4 and 5 at distance 1 -> {4,5}
Step 2: Merge 43 and 44 at distance 1 -> {43,44}
Step 3: $d(\{4,5\},0)=\max(4,5)=5$ -> merge {0,4,5}
Step 4: Merge 20 and 25 at distance 5 -> {20,25}
Step 5: $d(\{43,44\},39)=\max(4,5)=5$ -> merge {39,43,44}
Step 6: $d(\{20,25\},\{39,43,44\})=\max(19,23,24,14,18,19)=24$ -> merge
Step 7: $d(\{0,4,5\},\{20,25,39,43,44\})=\max \text{ pair distance} = |0-44| = 44$ -> final merge

Final answer: Merge distances: 1, 1, 5, 5, 5, 24, 44.

Interpretation: Complete linkage prefers compact clusters because it checks the farthest pair before merging.

Question 44

Topic focus: Average linkage hierarchical clustering

Question:

Using the same points, perform hierarchical agglomerative clustering using average linkage. Show the merge order.

Parameters used:

- Linkage = average linkage
- Cluster distance = mean of all pairwise distances

Detailed answer:

$d(A,B)$ = average of all pairwise distances between points in A and B

Step 1: Merge 4 and 5 at distance 1 -> {4,5}
Step 2: Merge 43 and 44 at distance 1 -> {43,44}
Step 3: $d(\{4,5\},0)=(4+5)/2=4.5$ -> merge {0,4,5}
Step 4: $d(\{43,44\},39)=(4+5)/2=4.5$ -> merge {39,43,44}
Step 5: Merge 20 and 25 at distance 5 -> {20,25}
Step 6: $d(\{0,4,5\},\{20,25\})=(20+25+16+21+15+20)/6 = 117/6 = 19.5$ -> merge {0,4,5,20,25}
Step 7: $d(\{39,43,44\},\{0,4,5,20,25\}) = \text{average of 15 pair distances} = 468/15 = 31.2$ -> final merge

Final answer: Merge distances: 1, 1, 4.5, 4.5, 5, 19.5, 31.2.

Interpretation: Average linkage balances single and complete linkage by considering overall pairwise distance.

Question 45

Topic focus: Compare linkage outcomes

Question:

Using the hierarchical clustering results for points 0,4,5,20,25,39,43,44, compare the behavior of single, complete, and average linkage.

Parameters used:

- Same data used for all methods
- Comparison basis = merge distances and cluster shape

Detailed answer:

Linkage	Rule	Effect on clustering
Single	Minimum pairwise distance	Merges by nearest connection; can chain clusters
Complete	Maximum pairwise distance	Requires all points to be relatively close; compact clusters
Average	Mean pairwise distance	Balanced compromise between nearest and farthest

Single linkage final merge happened at distance 15, much earlier than complete linkage final merge at 44. That shows single linkage connects clusters through nearest bridges.

Complete linkage delays merging if any points are very far apart. Average linkage sits between them.

Final answer: Single = chain-like, Complete = compact, Average = balanced.

Interpretation: Same data can produce different dendrograms depending on linkage rule.

Question 46

Topic focus: Small hierarchical clustering numerical

Question:

Consider one-dimensional points 1, 3, 8, and 12. Perform hierarchical clustering using single linkage and complete linkage. Show the merge order.

Parameters used:

- Points = 1,3,8,12
- Distance metric = absolute distance
- Linkages = single and complete

Detailed answer:

Pairwise distances: $d(1,3)=2$, $d(1,8)=7$, $d(1,12)=11$, $d(3,8)=5$, $d(3,12)=9$, $d(8,12)=4$.

Single linkage

Merge 1 and 3 at distance 2 $\rightarrow \{1,3\}$
Next $d(\{1,3\},8)=\min(7,5)=5$; $d(8,12)=4 \rightarrow$ merge 8 and 12 at distance 4
Final $d(\{1,3\},\{8,12\})=\min(7,11,5,9)=5 \rightarrow$ final merge

Complete linkage

Merge 1 and 3 at distance 2 $\rightarrow \{1,3\}$
Merge 8 and 12 at distance 4 $\rightarrow \{8,12\}$
Final $d(\{1,3\},\{8,12\})=\max(7,11,5,9)=11 \rightarrow$ final merge

Final answer: Single final merge distance = 5; complete final merge distance = 11.

Interpretation: Complete linkage keeps clusters compact and delays merging far-apart clusters.

Question 47

Topic focus: Dendrogram interpretation

Question:

A hierarchical clustering process gives the following merge distances: A and B merge at height 2, C and D merge at height 3, and cluster AB merges with cluster CD at height 8. Describe the dendrogram and identify clusters if it is cut at height 4.

Parameters used:

- Merge heights = 2,3,8
- Cut height = 4

Detailed answer:

The dendrogram has A and B joined at height 2. C and D are joined at height 3. The combined clusters AB and CD join only at height 8.

If we cut the dendrogram at height 4, any merge above 4 is not accepted.

AB has already formed because $2 < 4$. CD has already formed because $3 < 4$. AB and CD have not merged because $8 > 4$.

Final answer: Clusters at cut height 4 are {A,B} and {C,D}.

Interpretation: Cut height controls how many clusters are obtained from the dendrogram.

Question 48

Topic focus: Support and confidence

Question:

In 100 transactions, 40 contain bread, 30 contain butter, and 20 contain both bread and butter. Compute support of {bread,butter} and confidence of the rule bread $\rightarrow$ butter.

Parameters used:

- Total transactions = 100

- count(bread)=40
- count(butter)=30
- count(bread and butter)=20

Detailed answer:

$$\text{Support}(\{\text{bread}, \text{butter}\}) = \text{count}(\text{bread and butter}) / \text{total}$$

$$\text{Support} = 20/100 = 0.20$$

$$\text{Confidence}(\text{bread} \rightarrow \text{butter}) = \text{support}(\text{bread and butter}) / \text{support}(\text{bread})$$

$$\text{support}(\text{bread}) = 40/100 = 0.40$$

$$\text{Confidence} = 0.20/0.40 = 0.50$$

Final answer: Support = 0.20; Confidence = 0.50.

Interpretation: 50% of bread transactions also contain butter.

Question 49

Topic focus: Support, confidence, and lift

Question:

In 200 transactions, 80 contain milk, 60 contain bread, and 40 contain both milk and bread. Compute support, confidence of milk $\rightarrow$ bread, and lift.

Parameters used:

- Total = 200
- count(milk)=80
- count(bread)=60
- count(milk and bread)=40
- Rule = milk $\rightarrow$ bread

Detailed answer:

$$\text{support}(\text{milk and bread}) = 40/200 = 0.20$$

$$\text{support}(\text{milk}) = 80/200 = 0.40$$

$$\text{support}(\text{bread}) = 60/200 = 0.30$$

$$\text{confidence}(\text{milk} \rightarrow \text{bread}) = \text{support}(\text{milk and bread}) / \text{support}(\text{milk}) = 0.20/0.40 = 0.50$$

$$\text{lift} = \text{confidence} / \text{support}(\text{bread}) = 0.50/0.30 = 1.667$$

Final answer: Support = 0.20; Confidence = 0.50; Lift = 1.67.

Interpretation: Lift > 1 means milk and bread occur together more often than expected by chance.

Question 50

Topic focus: Apriori candidate pruning

Question:

Given transactions $T1=\{A,B,C\}$, $T2=\{A,B\}$, $T3=\{A,C\}$, $T4=\{B,C\}$, $T5=\{A,B,C\}$, and minimum support count = 3, find frequent 1-itemsets and frequent 2-itemsets.

Parameters used:

- Transactions = 5
- Minimum support count = 3
- Algorithm idea = Apriori

Detailed answer:

Frequent 1-itemsets

Count A: appears in $T1, T2, T3, T5 = 4$

Count B: appears in $T1, T2, T4, T5 = 4$

Count C: appears in $T1, T3, T4, T5 = 4$

All {A}, {B}, {C} are frequent because each count ≥ 3 .

Candidate 2-itemsets

Count {A,B}: appears in T1,T2,T5 = 3 -> frequent

Count {A,C}: appears in T1,T3,T5 = 3 -> frequent

Count {B,C}: appears in T1,T4,T5 = 3 -> frequent

Final answer: Frequent 1-itemsets: {A},{B},{C}. Frequent 2-itemsets: {A,B},{A,C},{B,C}.

Interpretation: Apriori keeps itemsets whose support count meets the minimum threshold.

Final Revision Priority

Practice in this order for fastest marks: normalization, Gini, precision/recall, rule coverage/accuracy, k-means, hierarchical clustering, and association rules. The longest and most error-prone are hierarchical clustering and k-means, so write them by hand at least twice before the exam.